

1. Questions

Study the following information carefully and answer the questions given below.

The given table shows the average of the total number of domestic and international shipments, and the number of domestic shipments handled by four different warehouses: Unit A, Unit B, Unit C, and Unit D in a month.

Warehouses	The average of the total number of domestic and international shipments	The number of domestic shipments
Unit A	600	700
Unit B	450	540
Unit C	800	960
Unit D	750	600

Note: The total number of shipments = The number of domestic shipments + The number of international shipments

Find the average of 35% of the total number of shipments for Unit C and the number of domestic shipments for Unit A.

- a. 630
- b. 730
- c. 560
- d. 640
- e. 620

2. Questions

What is the ratio of thrice the number of international shipments for Unit B to $\frac{2}{5}$ th of the number of domestic shipments for Unit D?

- a. 2:6
- b. 7:2
- c. 9:2
- d. 3:1
- e. 9:4

3. Questions

Find the difference between the total number of shipments for Unit A and 75% of one-half of the total number of shipments for Unit C.

- a. 700
- b. 650
- c. 800
- d. 500
- e. 600

4. Questions

A company charges Rs. x per domestic shipment and Rs. y per international shipment. If the total revenue generated by Unit A is Rs. 17,000 and Unit B is Rs. 12,600, find the value of y .

- a. 15
- b. 20
- c. 25
- d. 10
- e. 30

5. Questions

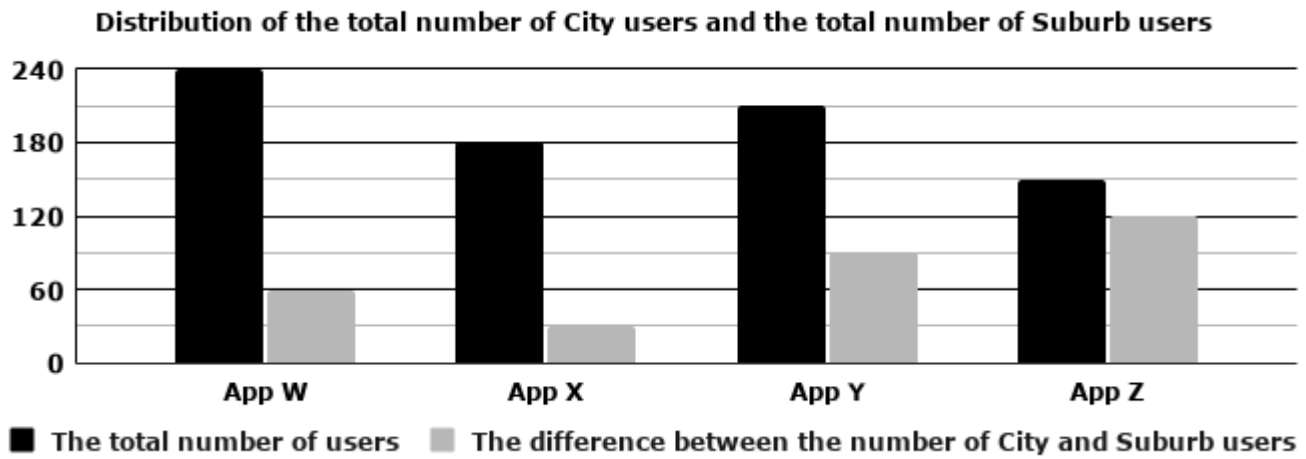
If Unit B adds x domestic shipments next month, making the new ratio of domestic to international shipments 7:4, find the value of x .

- a. 100
- b. 70
- c. 120
- d. 90
- e. 80

6. Questions

Study the following information carefully and answer the questions given below.

The given data represents the total number of users and the difference between the total number of City users and the total number of Suburb users for four different applications: App W, App X, App Y, and App Z in a day.



Note:

(i). The total number of City users is greater than the total number of Suburb users for all the given applications.

(ii). The total number of users = The total number of Suburb users + The total number of City users

If the total number of City users of App W is increased by 120, this new value is what percentage more than the total number of Suburb users of App W?

- a. 150%
- b. 300%
- c. 200%
- d. 100%
- e. 250%

7. Questions

Find the total sum of $\frac{3}{4}$ th of 80% of the total number of City users of App Z and 1.5 times the total number of Suburb users of App Y.

- a. 191
- b. 161
- c. 151
- d. 181
- e. 171

8. Questions

Calculate the average of half of the total number of users of App W and the total number of City users of App W decreased by 90.

- a. 90

- b. 120
- c. 80
- d. 100
- e. 110

9. Questions

App Y standard passes cost Rs. 100 more than premium passes. If all App Y Suburb users buy a premium pass and one-third of its City users buy a standard pass, the total revenue is Rs. 27000. Find the cost of 5 premium passes.

- a. Rs. 1300
- b. Rs. 1200
- c. Rs. 1100
- d. Rs. 1000
- e. Rs. 1500

10. Questions

App X imposes a fuel surcharge penalty on $y\%$ of its total number of Suburb users. If the app charges a penalty of Rs. 12 per affected user and collects a total penalty of Rs. 180, find the value of y .

- a. 30
- b. 20
- c. 40
- d. 15
- e. 25

11. Questions

Each of the given questions is followed by two statements I & II. You have to determine which statement(s) is/are sufficient/insufficient to answer the question and mark the answer accordingly.

A thief escapes from a bank and drives a car at a constant speed. A police officer starts chasing him after some time in a jeep. What is the speed of the thief?

I). The police officer drives at a speed of 15 km/hr and catches the thief 4 hours after starting the chase.

II). The thief started driving 2 hours before the police officer started the chase.

- a. The data given in both statements I and II together are not sufficient to answer the question.
- b. The data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.

- c. The data either in statement I alone or in statement II alone are sufficient to answer the question.
- d. The data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.
- e. The data in both statements I and II together are necessary to answer the question.

12. Questions

A and B started a business. A invested Rs. 5000 for 12 months. For how many months did B invest his money in the business?

I). The total profit earned at the end of the year from the business was Rs. 9000.

II). B invested Rs. 3000 more than A, and the profit share of A is 25% more than the profit share of B.

- a. The data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.
- b. The data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.
- c. The data given in both statements I and II together are not sufficient to answer the question.
- d. The data either in statement I alone or in statement II alone are sufficient to answer the question.
- e. The data in both statements I and II together are necessary to answer the question.

13. Questions

What is the total volume of a solid right circular cylinder?

I). The volume of a solid right circular cone having the same base radius and height as the given cylinder is 1232 cm^3 .

II). The total surface area of the solid right circular cylinder is 1760 cm^2 .

- a. The data in both statements I and II together are necessary to answer the question.
- b. The data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.
- c. The data given in both statements I and II together are not sufficient to answer the question.
- d. The data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.
- e. The data either in statement I alone or in statement II alone are sufficient to answer the question.

14. Questions

A vessel contains 120 liters of a milk and water in the ratio of 7:5, respectively. 'x' liters of mixture are replaced with water, then ratio becomes 7:17. This resulting mixture is then mixed with 'x' liters of milk. Find the final ratio of milk to water.

- a. 11:19
- b. 2:1
- c. 19:17
- d. 5:7
- e. 15:17

15. Questions

A, B, and C can complete a task in 12, 15, and 20 days. A and B start working together. After 3 days, A leaves and C joins B. B leaves x days before the completion of the task. If the task was completed in 10 days, find the value of x .

- a. 4
- b. 7
- c. 5
- d. 6
- e. 3

16. Questions

The simple interest earned on Rs. p at 10% p.a. for 3 years is reinvested at 20% p.a. compound interest. If the difference between the compound interest for the 3rd year and the 2nd year is Rs. 216, find the value of p .

- a. 18000
- b. 12000
- c. 15500
- d. 15000
- e. 10000

17. Questions

A, B, and C start a business with investments in the ratio of 3:4:5. Their investment times are $(y + 2)$ months, y months, and $(y - 2)$ months. The total annual profit is Rs. 34000 and C's profit share is Rs. 10000. Find the profit share of B.

- a. Rs. 12000
- b. Rs. 15000
- c. Rs. 10000
- d. Rs. 14000
- e. Rs. 12500

18. Questions

A shopkeeper marks his goods 40% above the cost price and offers a discount of 25%. However, while selling, he uses a weight of 800 grams instead of 1 kilogram. If he earns a profit of Rs. 425, find the cost price of 1 kilogram of the goods.

- a. Rs. 1260
- b. Rs. 1360
- c. Rs. 1700
- d. Rs. 1400
- e. Rs. 1350

19. Questions

The perimeter of a square is equal to the perimeter of a rectangle. The length of the rectangle is 20% more than its breadth. If the total area of the square is 36 square meters more than the total area of the rectangle, find the total cost of paving the rectangular floor at the rate of Rs. 5 per square meter.

- a. Rs. 21000
- b. Rs. 25000
- c. Rs. 22600
- d. Rs. 21780
- e. Rs. 21600

20. Questions

A man travels a total distance of 180 km. He covers some distance on a bike at 40 km/hr and the remaining distance in a car at 60 km/hr. If the distances traveled by each mode of transport is interchanged, the journey would take 30 minutes less. Find the total time taken to complete the original journey.

- a. 5 hours
- b. 6 hours
- c. 4 hours
- d. 3.5 hours
- e. 4.5 hours

21. Questions

What value will come in place of the question mark (?) in the following question?

$$3\frac{1}{4} * 16 + ?^2 - 12 = 4^3 + 25$$

- a. 17
- b. 7
- c. 3
- d. 9
- e. 13

22. Questions

$$?/\sqrt{144} + 18^2 - 45 = 12 * 25 - 15$$

- a. 62
- b. 82
- c. 92
- d. 72
- e. 52

23. Questions

$$(120 * 3)/5 + 6^3 - ?^2 = 14 * 5 + 22$$

- a. 34
- b. 12
- c. 14
- d. 16
- e. 24

24. Questions

$$5^3 - ? + 2\frac{1}{5} * 35 = 13^2 - 12$$

- a. 45
- b. 65
- c. 35
- d. 55
- e. 25

25. Questions

$$(\sqrt{1296}/?) * 15 + 21^2 - 86 = 445$$

- a. 4
- b. 16
- c. 12
- d. 8
- e. 6

26. Questions

What approximate value will come in place of the question mark (?) in the following question?

$$?^3 - 124.95 + (89.98/4.95) * 6.01 = 31.05^2 + 22.01$$

- a. 10
- b. 20
- c. 5
- d. 25
- e. 15

27. Questions

$$? + 15.05 * 11.98 - 8.01^3 = 2400.99/\sqrt{2401.05}$$

- a. 400
- b. 370
- c. 381
- d. 391
- e. 360

28. Questions

$$(\sqrt{2208.95/47.05}) * ? + 34.02^2 = 1354.01 - 85.99$$

- a. 102
- b. 95
- c. 130
- d. 122
- e. 112

29. Questions

64.98% of 820.05 + 23.01² - 115.99 = ? * 42.95

- a. 40
- b. 28
- c. 15
- d. 22
- e. 32

30. Questions

?/35.98 + 49.05² - 47.95² = 145.02 * 2.01

- a. 6900
- b. 6948
- c. 6958
- d. 7000
- e. 6800

31. Questions

What value will come in the place of the question mark (?) in the following number series?

10, 100, 25, 85, 45, ?, 70

- a. 75
- b. 45
- c. 55
- d. 60
- e. 65

32. Questions

48, 96, 240, ?, 2520, 10080

- a. 600
- b. 510
- c. 690
- d. 720
- e. 740

33. Questions

5, 9, 27, ?, 175, 355

- a. 65
- b. 75
- c. 85
- d. 95
- e. 70

34. Questions

3, 8, ?, 38, 78, 158

- a. 14
- b. 28
- c. 18
- d. 22
- e. 16

35. Questions

?, 15, 23, 40, 84, 209

- a. 8
- b. 12
- c. 5
- d. 10
- e. 20

36. Questions

Find the wrong term in the given series.

20, 32, 50, 72, 104, 140

- a. 140
- b. 32
- c. 72
- d. 104
- e. 50

37. Questions

4, 5, 12, 37, 160, 805

- a. 12
- b. 4
- c. 160
- d. 5
- e. 37

38. Questions

3, 15, 35, 63, 101, 143

- a. 15
- b. 63
- c. 143
- d. 101
- e. 35

39. Questions

50, 60, 40, 80, 0, 160, -120

- a. 160
- b. -120
- c. 0
- d. 60
- e. 40

40. Questions

240, 122, 67, 38.5, 27.25

- a. 67
- b. 240
- c. 27.25
- d. 122
- e. 38.5

41. Questions

In each of the following questions, two equations are given. You have to solve both the equations to find the relation between x and y.

I). $2x^2 - 15x + 27 = 0$

II). $2y^2 - 7y + 5 = 0$

- a. $x = y$ or the relationship cannot be established
- b. $x \leq y$
- c. $x \geq y$
- d. $x > y$
- e. $x < y$

42. Questions

I). $x^2 - 12x + 35 = 0$

II). $y^2 - 8y + 15 = 0$

- a. $x < y$
- b. $x \geq y$
- c. $x > y$
- d. $x = y$ or the relationship cannot be established
- e. $x \leq y$

43. Questions

I). $3x^2 + 16x + 21 = 0$

II). $y^2 + 3y + 2 = 0$

- a. $x < y$
- b. $x > y$
- c. $x \leq y$
- d. $x = y$ or the relationship cannot be established
- e. $x \geq y$

44. Questions

I). $x^2 + 10x + 24 = 0$

II). $2y^2 + 13y + 20 = 0$

- a. $x \geq y$
- b. $x < y$

- c. $x = y$ or the relationship cannot be established
- d. $x > y$
- e. $x \leq y$

45. Questions

I). $x^2 - 4x - 21 = 0$

II). $y^2 - 2y - 35 = 0$

- a. $x \leq y$
- b. $x \geq y$
- c. $x = y$ or the relationship cannot be established
- d. $x > y$
- e. $x < y$

Explanations:**1. Questions****Unit A**

The total number of shipments = $600 * 2 = 1200$

The number of domestic shipments = 700

The number of international shipments = $1200 - 700 = 500$

Unit B

The total number of shipments = $450 * 2 = 900$

The number of domestic shipments = 540

The number of international shipments = $900 - 540 = 360$

Unit C

The total number of shipments = $800 * 2 = 1600$

The number of domestic shipments = 960

The number of international shipments = $1600 - 960 = 640$

Unit D

The total number of shipments = $750 * 2 = 1500$

The number of domestic shipments = 600

The number of international shipments = $1500 - 600 = 900$

Warehouses	The total number of shipments	The number of domestic shipments	The number of international shipments
Unit A	1200	700	500
Unit B	900	540	360
Unit C	1600	960	640
Unit D	1500	600	900

Answer: A

The total number of shipments for Unit C = 1600

The 35% of the total number of shipments for Unit C = $(35/100) * 1600 = 560$

The number of domestic shipments for Unit A = 700

The required sum = $560 + 700 = 1260$

The average = $1260/2 = 630$

Hence, the correct answer is option A.

2. Questions

Unit A

The total number of shipments = $600 * 2 = 1200$

The number of domestic shipments = 700

The number of international shipments = $1200 - 700 = 500$

Unit B

The total number of shipments = $450 * 2 = 900$

The number of domestic shipments = 540

The number of international shipments = $900 - 540 = 360$

Unit C

The total number of shipments = $800 * 2 = 1600$

The number of domestic shipments = 960

The number of international shipments = $1600 - 960 = 640$

Unit D

The total number of shipments = $750 * 2 = 1500$

The number of domestic shipments = 600

The number of international shipments = $1500 - 600 = 900$

Warehouses	The total number of shipments	The number of domestic shipments	The number of international shipments
Unit A	1200	700	500
Unit B	900	540	360
Unit C	1600	960	640
Unit D	1500	600	900

Answer: C

The number of international shipments for Unit B = 360

Thrice the number of international shipments = $3 * 360 = 1080$

The number of domestic shipments for Unit D = 600

The $\frac{2}{5}$ th of The number of domestic shipments = $(\frac{2}{5}) * 600 = 240$

The required ratio = $1080:240 = 9:2$

Hence, the correct answer is option C.

3. Questions

Unit A

The total number of shipments = $600 * 2 = 1200$

The number of domestic shipments = 700

The number of international shipments = $1200 - 700 = 500$

Unit B

The total number of shipments = $450 * 2 = 900$

The number of domestic shipments = 540

The number of international shipments = $900 - 540 = 360$

Unit C

The total number of shipments = $800 * 2 = 1600$

The number of domestic shipments = 960

The number of international shipments = $1600 - 960 = 640$

Unit D

The total number of shipments = $750 * 2 = 1500$

The number of domestic shipments = 600

The number of international shipments = $1500 - 600 = 900$

Warehouses	The total number of shipments	The number of domestic shipments	The number of international shipments
Unit A	1200	700	500
Unit B	900	540	360
Unit C	1600	960	640
Unit D	1500	600	900

Answer: E

The total number of shipments for Unit C = 1600

The one-half of the total number of shipments = $(1/2) * 1600 = 800$

The 75% of this value = $(75/100) * 800 = 600$

The total number of shipments for Unit A = 1200

The required difference = $1200 - 600 = 600$

Hence, the correct answer is option E.

4. Questions

Unit A

The total number of shipments = $600 * 2 = 1200$

The number of domestic shipments = 700

The number of international shipments = $1200 - 700 = 500$

Unit B

The total number of shipments = $450 * 2 = 900$

The number of domestic shipments = 540

The number of international shipments = $900 - 540 = 360$

Unit C

The total number of shipments = $800 * 2 = 1600$

The number of domestic shipments = 960

The number of international shipments = $1600 - 960 = 640$

Unit D

The total number of shipments = $750 * 2 = 1500$

The number of domestic shipments = 600

The number of international shipments = $1500 - 600 = 900$

Warehouses	The total number of shipments	The number of domestic shipments	The number of international shipments
Unit A	1200	700	500
Unit B	900	540	360
Unit C	1600	960	640
Unit D	1500	600	900

Answer: B

The number of domestic shipments for Unit A = 700

The number of international shipments for Unit A = 500

$$700x + 500y = 17000$$

$$7x + 5y = 170 \text{ --- (1)}$$

The number of domestic shipments for Unit B = 540

The number of international shipments for Unit B = 360

$$540x + 360y = 12600$$

$$3x + 2y = 70 \text{ --- (2)}$$

Multiply Equation (2) by 2.5,

$$7.5x + 5y = 175 \text{ --- (3)}$$

Subtract Equation (1) from (3):

$$(7.5x + 5y) - (7x + 5y) = 175 - 170$$

$$0.5x = 5$$

$$x = 10$$

Substitute the value of x into Equation (2),

$$3(10) + 2y = 70$$

$$30 + 2y = 70$$

$$2y = 40$$

$$y = 20$$

Hence, the correct answer is option B.

5. Questions

Unit A

The total number of shipments = $600 * 2 = 1200$

The number of domestic shipments = 700

The number of international shipments = $1200 - 700 = 500$

Unit B

The total number of shipments = $450 * 2 = 900$

The number of domestic shipments = 540

The number of international shipments = $900 - 540 = 360$

Unit C

The total number of shipments = $800 * 2 = 1600$

The number of domestic shipments = 960

The number of international shipments = $1600 - 960 = 640$

Unit D

The total number of shipments = $750 * 2 = 1500$

The number of domestic shipments = 600

The number of international shipments = $1500 - 600 = 900$

Warehouses	The total number of shipments	The number of domestic shipments	The number of international shipments
Unit A	1200	700	500
Unit B	900	540	360
Unit C	1600	960	640
Unit D	1500	600	900

Answer: D

The initial total number of international shipments for Unit B = 360

4 ratio parts = 360.

The 1 ratio part = $360/4 = 90$

The new total number of domestic shipments = $7 * 90 = 630$

The new total number of shipments = $630 + 360 = 990$

The initial total number of shipments for Unit B = 900

The value of $x = 990 - 900 = 90$

Hence, the correct answer is option D.

6. Questions

App W

The total number of City users = $(240 + 60)/2 = 150$

The total number of Suburb users = $(240 - 60)/2 = 90$

The total number of users = 240

App X

The total number of City users = $(180 + 30)/2 = 105$

The total number of Suburb users = $(180 - 30)/2 = 75$

The total number of users = 180

App Y

The total number of City users = $(210 + 90)/2 = 150$

The total number of Suburb users = $(210 - 90)/2 = 60$

The total number of users = 210

App Z

The total number of City users = $(150 + 120)/2 = 135$

The total number of Suburb users = $(150 - 120)/2 = 15$

The total number of users = 150

The name of the Application	The total number of users	The total number of City users	The total number of Suburb users
App W	240	150	90
App X	180	105	75
App Y	210	150	60
App Z	150	135	15

Answer: C

The total number of City users of App W = 150

The value after being increased by 120 = $150 + 120 = 270$

The total number of Suburb users of App W = 90

The difference = $270 - 90 = 180$

The required percentage = $(180/90) * 100 = 2 * 100 = 200\%$

Hence, the correct answer is option C.

7. Questions

App W

The total number of City users = $(240 + 60)/2 = 150$

The total number of Suburb users = $(240 - 60)/2 = 90$

The total number of users = 240

App X

The total number of City users = $(180 + 30)/2 = 105$

The total number of Suburb users = $(180 - 30)/2 = 75$

The total number of users = 180

App Y

The total number of City users = $(210 + 90)/2 = 150$

The total number of Suburb users = $(210 - 90)/2 = 60$

The total number of users = 210

App Z

The total number of City users = $(150 + 120)/2 = 135$

The total number of Suburb users = $(150 - 120)/2 = 15$

The total number of users = 150

The name of the Application	The total number of users	The total number of City users	The total number of Suburb users
App W	240	150	90
App X	180	105	75
App Y	210	150	60
App Z	150	135	15

Answer: E

The total number of City users of App Z = 135

The 80% of the total number of City users = $(80/100) * 135 = 108$

The 3/4th of this intermediate value = $(3/4) * 108 = 81$

The total number of Suburb users of App Y = 60

The 1.5 times the total number of Suburb users = $1.5 * 60 = 90$

The required sum = $81 + 90 = 171$

Hence, the correct answer is option E.

8. Questions

App W

The total number of City users = $(240 + 60)/2 = 150$

The total number of Suburb users = $(240 - 60)/2 = 90$

The total number of users = 240

App X

The total number of City users = $(180 + 30)/2 = 105$

The total number of Suburb users = $(180 - 30)/2 = 75$

The total number of users = 180

App Y

The total number of City users = $(210 + 90)/2 = 150$

The total number of Suburb users = $(210 - 90)/2 = 60$

The total number of users = 210

App Z

The total number of City users = $(150 + 120)/2 = 135$

The total number of Suburb users = $(150 - 120)/2 = 15$

The total number of users = 150

The name of the Application	The total number of users	The total number of City users	The total number of Suburb users
App W	240	150	90
App X	180	105	75
App Y	210	150	60
App Z	150	135	15

Answer: A

The total number of users of App W = 240

The half of this value = $(1/2) * 240 = 120$

The total number of City users of App W = 150

This value decreased by 90 = $150 - 90 = 60$

The required sum = $120 + 60 = 180$

The average = $180/2 = 90$

Hence, the correct answer is option A.

9. Questions

App W

The total number of City users = $(240 + 60)/2 = 150$

The total number of Suburb users = $(240 - 60)/2 = 90$

The total number of users = 240

App X

The total number of City users = $(180 + 30)/2 = 105$

The total number of Suburb users = $(180 - 30)/2 = 75$

The total number of users = 180

App Y

The total number of City users = $(210 + 90)/2 = 150$

The total number of Suburb users = $(210 - 90)/2 = 60$

The total number of users = 210

App Z

The total number of City users = $(150 + 120)/2 = 135$

The total number of Suburb users = $(150 - 120)/2 = 15$

The total number of users = 150

The name of the Application	The total number of users	The total number of City users	The total number of Suburb users
App W	240	150	90
App X	180	105	75
App Y	210	150	60
App Z	150	135	15

Answer: D

The total number of Suburb users of App Y = 60

The total number of City users of App Y = 150

The number of City users who purchased the standard pass = $(1/3) * 150 = 50$

Let the cost of 1 premium pass be p.

Then, the cost of 1 standard pass = $p + 100$

The total revenue = (Suburb users * cost of premium pass) + (Standard pass buyers * cost of standard pass)

$$27000 = 60 * p + 50 * (p + 100)$$

$$27000 = 60p + 50p + 5000$$

$$27000 = 110p + 5000$$

$$22000 = 110p$$

$$p = 200$$

The cost of 1 premium pass = Rs. 200

The total cost of 5 premium passes = $5 * 200 = 1000$

Hence, the correct answer is option D.

10. Questions

App W

The total number of City users = $(240 + 60)/2 = 150$

The total number of Suburb users = $(240 - 60)/2 = 90$

The total number of users = 240

App X

The total number of City users = $(180 + 30)/2 = 105$

The total number of Suburb users = $(180 - 30)/2 = 75$

The total number of users = 180

App Y

The total number of City users = $(210 + 90)/2 = 150$

The total number of Suburb users = $(210 - 90)/2 = 60$

The total number of users = 210

App Z

The total number of City users = $(150 + 120)/2 = 135$

The total number of Suburb users = $(150 - 120)/2 = 15$

The total number of users = 150

The name of the Application	The total number of users	The total number of City users	The total number of Suburb users
App W	240	150	90
App X	180	105	75
App Y	210	150	60
App Z	150	135	15

Answer: B

The actual magnitude of the total number of Suburb users of App X = 75

The number of users affected by the penalty = $y\%$ of 75 = $(y/100) * 75 = 0.75y$

The penalty charged per affected user = Rs. 12

The total penalty collected = $0.75y * 12 = 9y$

$9y = 180$

$y = 180/9 = 20$

Hence, the correct answer is option B.

11. Questions

Answer: E

The formula for distance is $\text{Distance} = \text{Speed} * \text{Time}$.

From statement I:

The speed of the police officer = 15 km/hr

The time taken by the police officer to catch the thief = 4 hours

The total distance traveled by the police officer = $15 * 4 = 60$ km.

Since the police officer catches the thief, the total distance traveled by the thief is also 60 km. However, the time taken by the thief is unknown, so the speed cannot be calculated. Thus, statement I alone is not sufficient.

From statement II:

The thief started 2 hours before the police officer.

No distance or speed values are given. Thus, statement II alone is not sufficient.

From both statements I and II together:

The total distance traveled by the thief = 60 km.

The total time taken by the thief = The time taken by the police officer + 2 hours = $4 + 2 = 6$ hours.

The speed of the thief = $60/6 = 10$ km/hr.

Therefore, the data in both statements I and II together are necessary to answer the question.

Hence, the correct answer is option E.

12. Questions

Answer: B

From statement I:

The total profit = Rs. 9000.

There is no information regarding the investment amount of B or the individual profit shares. Thus, statement I alone is not sufficient.

From statement II:

The investment of B = The investment of A + 3000 = $5000 + 3000 =$ Rs. 8000.

The profit share of A = The profit share of B + 25% of the profit share of B = The profit share of B + $(25/100) * \text{The profit share of B} = 1.25 * \text{The profit share of B}$.

The ratio of the profit share of A to B = $1.25:1 = 125:100 = 5:4$.

Let the time period for B be t months.

The ratio of profit shares is equal to the ratio of the product of investment and time.

$$(5000 * 12)/(8000 * t) = 5/4$$

$$60000/(8000 * t) = 5/4$$

$$60/(8 * t) = 5/4$$

$$240 = 40 * t$$

$$t = 6 \text{ months.}$$

Therefore, the data in statement II alone are sufficient to answer the question.

Hence, the correct answer is option B.

13. Questions

Answer: D

$$\text{The volume of a cone} = (1/3) * (22/7) * r^2 * h.$$

$$\text{The volume of a cylinder} = (22/7) * r^2 * h.$$

$$\text{The total surface area of a cylinder} = 2 * (22/7) * r * (r + h).$$

From statement I:

$$\text{The volume of a cone} = (1/3) * (22/7) * r^2 * h = 1232 \text{ cm}^3.$$

$$\text{Therefore, the volume of the cylinder} = 3 * \text{The volume of the cone.}$$

$$\text{The volume of the cylinder} = 3 * 1232 = 3696 \text{ cm}^3.$$

Thus, statement I alone is sufficient.

From statement II:

$$\text{The total surface area of the cylinder} = 2 * (22/7) * r * (r + h) = 1760 \text{ cm}^2.$$

Here, we have one equation with two unknown variables (r and h). It is impossible to determine a unique value for the volume without knowing either the radius or the height. Thus, statement II alone is not sufficient.

Therefore, the data in statement I alone are sufficient to answer the question.

Hence, the correct answer is option D.

14. Questions

Answer: C

$$\text{Let the total initial quantity of the mixture} = 120 \text{ L}$$

$$\text{The initial quantity of milk} = (7/12) * 120 = 70 \text{ L}$$

$$\text{The initial quantity of water} = (5/12) * 120 = 50 \text{ L}$$

$$\text{When } x \text{ liters of the mixture is removed, the quantity of milk removed} = (7/12) * x$$

$$\text{The quantity of milk after replacement} = 70 - (7/12) * x$$

According to the question, the ratio of milk to water after replacement is 7:17

The total parts in the vessel = $7 + 17 = 24$

The quantity of milk after replacement = $(7/24) * 120 = 35$ L

$70 - (7/12) * x = 35$

$(7/12) * x = 35$

$x = (35 * 12)/7 = 60$

The quantity of water after replacement = $120 - 35 = 85$ L

According to the question, the mixture is then mixed with x liters of milk

The quantity of milk added = 60 L

The final quantity of milk = $35 + 60 = 95$ L

The final quantity of water = 85 L

The final ratio of milk to water = $95:85 = 19:17$

Hence, the correct answer is option C.

15. Questions

Answer: A

The total work units = LCM of 12, 15, and 20 = 60 units

The daily efficiency of A = $60/12 = 5$ units

The daily efficiency of B = $60/15 = 4$ units

The daily efficiency of C = $60/20 = 3$ units

The work done by A and B in the first 3 days = $(5 + 4) * 3 = 27$ units

The remaining work after 3 days = $60 - 27 = 33$ units

The remaining time to complete the 10-day task = $10 - 3 = 7$ days

C works for the entire remaining 7 days.

The work done by C = $7 * 3 = 21$ units

The remaining work done by B in the second phase = $33 - 21 = 12$ units

The number of days B worked in the second phase = $12/4 = 3$ days

The total days B worked = $3 + 3 = 6$ days

The number of days before completion that B left (x) = $10 - 6 = 4$

Hence, the correct answer is option A.

16. Questions

Answer: D

Simple interest = (Principal * Rate * Time)/100.

The total simple interest earned after 3 years = $(p * 10 * 3)/100 = 30\%$ of $p = (30/100) * p = 0.3 * p$

The principal for the compound interest scheme = $0.3 * p$

The amount at the end of the 1st year in CI = 120% of $(0.3 * p) = (120/100) * 0.3 * p = 0.36 * p$

The compound interest for the 2nd year = 20% of $(0.36 * p) = (20/100) * 0.36 * p = 0.072 * p$

The amount at the end of the 2nd year in CI = 120% of $(0.36 * p) = (120/100) * 0.36 * p = 0.432 * p$

The compound interest for the 3rd year = 20% of $(0.432 * p) = (20/100) * 0.432 * p = 0.0864 * p$

The difference between the 3rd year and 2nd year interest = $0.0864 * p - 0.072 * p = 0.0144 * p$

$0.0144 * p = 216$

$p = 216/0.0144 = 15000$

Hence, the correct answer is option D.

17. Questions

Answer: A

The effective investment weight of A = $3 * (y + 2) = 3y + 6$

The effective investment weight of B = $4 * y = 4y$

The effective investment weight of C = $5 * (y - 2) = 5y - 10$

The total effective investment weight of all partners = $3y + 6 + 4y + 5y - 10 = 12y - 4$

The ratio of C's profit to the total profit = $(5y - 10)/(12y - 4)$

$(5y - 10)/(12y - 4) = 10000/34000$

$(5y - 10)/(12y - 4) = 5/17$

$17 * (5y - 10) = 5 * (12y - 4)$

$85y - 170 = 60y - 20$

$25y = 150$

$y = 6$

The effective investment weight of B = $4 * 6 = 24$

The total effective investment weight = $12 * 6 - 4 = 68$

The profit share of B = $(24/68) * 34000 = \text{Rs. } 12000$

Hence, the correct answer is option A.

18. Questions

Answer: C

Profit percentage = $((\text{Selling Price} - \text{Cost Price})/\text{Cost Price}) * 100$.

Let the actual cost price of 1 gram of goods be 1 unit.

The original cost price of the goods = 1000 units.

The cost price of the delivered goods = 800 units.

The customer believes they are receiving 1000 grams. The marked price of 1000 grams = 140% of 1000 = $(140/100) * 1000 = 1400$ units.

The scheme is a discount of 25%.

The final selling price = 75% of 1400 = $(75/100) * 1400 = 1050$ units.

The total profit generated = 1050 - 800 = 250 units.

250 units = 425

1 unit = $425/250 = 1.7$

The original cost price = $1000 * 1.7 = \text{Rs. } 1700$

Hence, the correct answer is option C.

19. Questions

Answer: E

The perimeter of a rectangle = $2 * (\text{Length} + \text{Breadth})$.

The perimeter of a square = $4 * \text{Side}$.

The area of a square = Side^2 .

The area of a rectangle = $\text{Length} * \text{Breadth}$.

Let the breadth of the rectangle be $5k$.

The length of the rectangle = $5k + 20\% \text{ of } 5k = 5k + (20/100) * 5k = 6k$

The perimeter of the rectangle = $2 * (6k + 5k) = 22k$

The perimeter of the square = $22k$

The side of the square = $22k/4 = 5.5k$

The total area of the square = $5.5k * 5.5k = 30.25k^2$

The total area of the rectangle = $6k * 5k = 30k^2$

The difference in area = $30.25k^2 - 30k^2 = 0.25k^2$

$0.25k^2 = 36$

$k^2 = 36/0.25 = 144$

The total area of the rectangle = $30 * 144 = 4320 \text{ m}^2$

The total cost of paving the rectangular floor = $4320 * 5 = \text{Rs. } 21600$

Hence, the correct answer is option E.

20. Questions

Answer: C

Time = Distance/Speed.

Let the distance traveled by bike be d km.

The distance traveled by car = 180 - d km

The time taken for the original journey = $d/40 + (180 - d)/60$

The time taken for the interchanged journey = $(180 - d)/40 + d/60$

The difference in time is 30 minutes, which is 0.5 hours.

$$(d/40 + (180 - d)/60) - ((180 - d)/40 + d/60) = 0.5$$

$$d/40 - d/60 - (180 - d)/40 + (180 - d)/60 = 0.5$$

$$d/120 - (180 - d)/120 = 0.5$$

$$(2d - 180)/120 = 0.5$$

$$2d - 180 = 60$$

$$2d = 240$$

$$d = 120$$

The original time taken = $120/40 + (180 - 120)/60 = 3 + 1 = 4$ hours

Hence, the correct answer is option C.

21. Questions

Answer: B

$$3\frac{1}{4} * 16 + ?^2 - 12 = 4^3 + 25$$

$$13/4 * 16 + ?^2 - 12 = 64 + 25$$

$$52 + ?^2 - 12 = 89$$

$$40 + ?^2 = 89$$

$$?^2 = 89 - 40$$

$$?^2 = 49$$

$$? = 7$$

Hence, the correct answer is option B.

22. Questions

Answer: D

$$\sqrt{144 + 18^2} - 45 = 12 * 25 - 15$$

$$\sqrt{12 + 324} - 45 = 300 - 15$$

$$\sqrt{12 + 279} = 285$$

$$\sqrt{12} = 285 - 279$$

$$\sqrt{12} = 6$$

$$? = 6 * 12$$

$$? = 72$$

Hence, the correct answer is option D.

23. Questions

Answer: C

$$(120 * 3)/5 + 6^3 - ?^2 = 14 * 5 + 22$$

$$(120 * 3)/5 + 216 - ?^2 = 70 + 22$$

$$72 + 216 - ?^2 = 92$$

$$288 - ?^2 = 92$$

$$?^2 = 288 - 92$$

$$?^2 = 196$$

$$? = 14$$

Hence, the correct answer is option C.

24. Questions

Answer: A

$$5^3 - ? + 2\frac{1}{5} * 35 = 13^2 - 12$$

$$125 - ? + 11/5 * 35 = 169 - 12$$

$$125 - ? + 77 = 157$$

$$202 - ? = 157$$

$$? = 202 - 157$$

$$? = 45$$

Hence, the correct answer is option A.

25. Questions

Answer: E

$$(\sqrt{1296/?}) * 15 + 21^2 - 86 = 445$$

$$(36/?)*15 + 441 - 86 = 445$$

$$540/? + 355 = 445$$

$$540/? = 445 - 355$$

$$540/? = 90$$

$$? = 540/90$$

$$? = 6$$

Hence, the correct answer is option E.

26. Questions

Answer: A

$$?^3 - 124.95 + (89.98/4.95) * 6.01 = 31.05^2 + 22.01$$

$$?^3 - 125 + (90/5) * 6 = 31^2 + 22$$

$$?^3 - 125 + 18 * 6 = 961 + 22$$

$$?^3 - 125 + 108 = 983$$

$$?^3 - 17 = 983$$

$$?^3 = 983 + 17$$

$$?^3 = 1000$$

$$? = 10$$

Hence, the correct answer is option A.

27. Questions

Answer: C

$$? + 15.05 * 11.98 - 8.013 = 2400.99/\sqrt{2401.05}$$

$$? + 15 * 12 - 8^3 = 2401/\sqrt{2401}$$

$$? + 180 - 512 = 2401/49$$

$$? - 332 = 49$$

$$? = 332 + 49$$

$$? = 381$$

Hence, the correct answer is option C.

28. Questions**Answer: E**

$$(\sqrt{2208.95/47.05}) * ? + 34.02^2 = 1354.01 - 85.99$$

$$(\sqrt{2209/47}) * ? + 34^2 = 1354 - 86$$

$$(47/47) * ? + 1156 = 1268$$

$$1 * ? + 1156 = 1268$$

$$? = 1268 - 1156$$

$$? = 112$$

Hence, the correct answer is option E.

29. Questions**Answer: D**

$$64.98\% \text{ of } 820.05 + 23.01^2 - 115.99 = ? * 42.95$$

$$65\% \text{ of } 820 + 23^2 - 116 = ? * 43$$

$$(65/100) * 820 + 529 - 116 = ? * 43$$

$$533 + 529 - 116 = ? * 43$$

$$1062 - 116 = ? * 43$$

$$946 = ? * 43$$

$$? = 946/43$$

$$? = 22$$

Hence, the correct answer is option D.

30. Questions**Answer: B**

$$?/35.98 + 49.05^2 - 47.95^2 = 145.02 * 2.01$$

$$?/36 + 49^2 - 48^2 = 145 * 2$$

$$?/36 + 2401 - 2304 = 290$$

$$?/36 + 97 = 290$$

$$?/36 = 290 - 97$$

$$?/36 = 193$$

$$? = 193 * 36$$

$$? = 6948$$

Hence, the correct answer is option B.

31. Questions

Answer: E

The first series:

$$10 + 15 = 25$$

$$25 + 20 = 45$$

$$45 + 25 = 70$$

The second series:

$$100 - 15 = 85$$

$$85 - 20 = \mathbf{65}$$

Hence, the correct answer is option E.

32. Questions

Answer: D

$$48 * 2 = 96$$

$$96 * 2.5 = 240$$

$$240 * 3 = \mathbf{720}$$

$$720 * 3.5 = 2520$$

$$2520 * 4 = 10080$$

Hence, the correct answer is option D.

33. Questions

Answer: B

$$5 + (2^3 - 2^2) = 5 + 4 = 9$$

$$9 + (3^3 - 3^2) = 9 + 18 = 27$$

$$27 + (4^3 - 4^2) = 27 + 48 = \mathbf{75}$$

$$75 + (5^3 - 5^2) = 75 + 100 = 175$$

$$175 + (6^3 - 6^2) = 175 + 180 = 355$$

Hence, the correct answer is option B.

34. Questions

Answer: C

$$(3 + 1) * 2 = 8$$

$$(8 + 1) * 2 = 18$$

$$(18 + 1) * 2 = 38$$

$$(38 + 1) * 2 = 78$$

$$(78 + 1) * 2 = 158$$

Hence, the correct answer is option C.

35. Questions

Answer: D

10	15	23	40	84	209
	+5	+8	+17	+44	+125
		+3 ¹	+3 ²	+3 ³	+3 ⁴

Hence, the correct answer is option D.

36. Questions

Answer: C

$$20 + (6 * 2) = 32$$

$$32 + (6 * 3) = 50$$

$$50 + (6 * 4) = 74$$

$$74 + (6 * 5) = 104$$

$$104 + (6 * 6) = 140$$

Hence, the correct answer is option C.

37. Questions

Answer: E

$$4 * 1 + 1 = 5$$

$$5 * 2 + 2 = 12$$

$$12 * 3 + 3 = 39$$

$$39 * 4 + 4 = 160$$

$$160 * 5 + 5 = 805$$

Hence, the correct answer is option E.

38. Questions

Answer: D

$$2^2 - 1 = 3$$

$$4^2 - 1 = 15$$

$$6^2 - 1 = 35$$

$$8^2 - 1 = 63$$

$$10^2 - 1 = \mathbf{99}$$

$$12^2 - 1 = 143$$

Hence, the correct answer is option D.

39. Questions

Answer: B

50	60	40	80	0	160	-160
+10	-20	+40	-80	+160	-320	

Hence, the correct answer is option B.

40. Questions

Answer: A

$$240/2 + 2 = 122$$

$$122/2 + 4 = \mathbf{65}$$

$$65/2 + 6 = 38.5$$

$$38.5/2 + 8 = 27.25$$

Hence, the correct answer is option A.

41. Questions

Answer: D

I). $2x^2 - 15x + 27 = 0$

$$2x^2 - 9x - 6x + 27 = 0$$

$$x(2x - 9) - 3(2x - 9) = 0$$

$$(x - 3)(2x - 9) = 0$$

$$x = 3, 4.5$$

II). $2y^2 - 7y + 5 = 0$

$$2y^2 - 5y - 2y + 5 = 0$$

$$y(2y - 5) - 1(2y - 5) = 0$$

$$(y - 1)(2y - 5) = 0$$

$$y = 1, 2.5$$

Comparison:

$$3 > 1$$

$$3 > 2.5$$

$$4.5 > 1$$

$$4.5 > 2.5$$

Therefore, $x > y$.

Hence, the correct answer is option D.

42. Questions

Answer: B

I). $x^2 - 12x + 35 = 0$

$$x^2 - 7x - 5x + 35 = 0$$

$$x(x - 7) - 5(x - 7) = 0$$

$$(x - 5)(x - 7) = 0$$

$$x = 5, 7$$

II). $y^2 - 8y + 15 = 0$

$$y^2 - 5y - 3y + 15 = 0$$

$$y(y - 5) - 3(y - 5) = 0$$

$$(y - 3)(y - 5) = 0$$

$$y = 3, 5$$

Comparison:

$$7 > 5$$

$$7 > 3$$

$$5 = 5$$

$$5 > 3$$

Therefore, $x \geq y$.

Hence, the correct answer is option B.

43. Questions

Answer: A

I). $3x^2 + 16x + 21 = 0$

$$3x^2 + 9x + 7x + 21 = 0$$

$$3x(x + 3) + 7(x + 3) = 0$$

$$(3x + 7)(x + 3) = 0$$

$$x = -3, -2.33$$

$$\text{II). } y^2 + 3y + 2 = 0$$

$$y^2 + 2y + 1y + 2 = 0$$

$$y(y + 2) + 1(y + 2) = 0$$

$$(y + 1)(y + 2) = 0$$

$$y = -1, -2$$

Comparison:

$$-3 < -2$$

$$-3 < -1$$

$$-2.33 < -2$$

$$-2.33 < -1$$

Therefore, $x < y$.

Hence, the correct answer is option A.

44. Questions

Answer: E

$$\text{I). } x^2 + 10x + 24 = 0$$

$$x^2 + 6x + 4x + 24 = 0$$

$$x(x + 6) + 4(x + 6) = 0$$

$$(x + 4)(x + 6) = 0$$

$$x = -4, -6$$

$$\text{II). } 2y^2 + 13y + 20 = 0$$

$$2y^2 + 8y + 5y + 20 = 0$$

$$2y(y + 4) + 5(y + 4) = 0$$

$$(2y + 5)(y + 4) = 0$$

$$y = -4, -2.5$$

Comparison:

$$-6 < -4$$

$$-6 < -2.5$$

$$-4 = -4$$

$$-4 < -2.5$$

Therefore, $x \leq y$.

Hence, the correct answer is option E.

45. Questions

Answer: C

I). $x^2 - 4x - 21 = 0$

$$x^2 - 7x + 3x - 21 = 0$$

$$x(x - 7) + 3(x - 7) = 0$$

$$(x + 3)(x - 7) = 0$$

$$x = 7, -3$$

II). $y^2 - 2y - 35 = 0$

$$y^2 - 7y + 5y - 35 = 0$$

$$y(y - 7) + 5(y - 7) = 0$$

$$(y + 5)(y - 7) = 0$$

$$y = 7, -5$$

Comparison:

$$7 = 7$$

$$7 > -5$$

$$-3 < 7$$

$$-3 > -5$$

Therefore, $x = y$ or the relationship cannot be established.

Hence, the correct answer is option C.